WFH Burnout Predictor: A CRISP-DM Machine Learning Project

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***Abstract*—This paper describes a machine learning project for estimating burnout in work-from-home settings from daily behavioral indicators such as working hours, screen time, meetings, breaks, after-hours work, sleep and task completion. The work follows the CRISP-DM methodology discussed in the course The main task is regression on `burnout_score`; classification of `burnout_risk` is treated as a secondary analysis because the high-risk class is strongly imbalanced.**

***Index Terms*—machine learning, CRISP-DM, burnout prediction, work from home, regression, classification, fairness**

I. Introduction

The goal of the project is to develop a complete machine learning workflow for the prediction of employee burnout in work-from-home (WFH) scenarios. Burnout is a sensitive topic: predictions should not be interpreted as medical diagnoses or used for automatic human-resources decisions. The intended use is decision support at an aggregate level, for example to identify workload patterns that may deserve organizational attention.

The project uses the public Kaggle dataset *Work From Home Employee Burnout Dataset* [1]. It contains 1,800 daily observations, 180 users and 10 records per user. Each row includes workload and lifestyle variables: `day_type`, `work_hours`, `screen_time_hours`, `meetings_count`, `breaks_taken`, `after_hours_work`, `sleep_hours` and `task_completion_rate`. The available targets are `burnout_score`, a continuous score, and `burnout_risk`, a categorical label with values `Low`, `Medium` and `High`.

The work was carried out individually by Simone Mastria and covers dataset inspection, preprocessing design, feature engineering, model selection, evaluation and interpretation.

The main challenge is not the implementation of a single algorithm, but the construction of a defensible data mining workflow. In particular, the dataset contains repeated observations for each employee. If a standard random row split were used, the same employee could appear in both training and test data, producing an overly optimistic evaluation. Another challenge is the semantic interpretation of the available variables: some features are plausible behavioral predictors, while others may be close proxies for the target itself. These aspects motivated a conservative methodology based on grouped splitting, baseline comparison and explicit discussion of limitations.

II. Proposed Method

A. CRISP-DM Strategy

The project was organized according to CRISP-DM [2]. In the business understanding phase, the objective was formulated as a predictive and interpretable support tool for WFH burnout monitoring. In the data understanding phase, the dataset schema, missing values, duplicates, target distributions and correlations were analyzed. In the data preparation phase, transformations were implemented through scikit-learn pipelines to avoid leakage. Modeling compared baselines and supervised learning algorithms. Evaluation considered both technical metrics and domain limitations. Deployment was described as a cautious plan rather than an automatic production system.

Table I summarizes how the CRISP-DM phases were mapped to the project artifacts. This mapping was used both to structure the notebook and to prepare the final paper.

TABLE I
CRISP-DM mapping for the project.

Phase	Project artifact
Business understanding	Goal, target choice, metrics and risks
Data understanding	Schema, target distribution, correlations and bias notes
Data preparation	Grouped split, encoding, scaling and feature engineering
Modeling	Baselines, candidate models and tuning
Evaluation	Test metrics, error analysis and interpretability
Deployment	Colab execution, saved pipeline and monitoring plan

B. Data Preparation

The dataset has no missing values and no duplicated rows. However, it still requires preprocessing, The identifier `user_id` was not used as a feature because it is not generalizable. Instead, it was used as a grouping variable during splitting: records from the same user must not appear in both training and test sets.

Table II reports the core variables. The distinction between predictive features, grouping variable and targets is essential to avoid leakage.

The train/test split was performed with `GroupShuffleSplit`, using 80% of the records for

TABLE II
MAIN VARIABLES USED IN THE PROJECT.

Variable	Role
user_id	Grouping variable, excluded from predictors
day_type	Categorical predictor
work_hours	Numeric predictor
screen_time_hours	Numeric predictor
meetings_count	Numeric predictor
breaks_taken	Numeric predictor
after_hours_work	Binary predictor
sleep_hours	Numeric predictor
task_completion_rate	Numeric predictor, high-leverage feature
burnout_score	Regression target
burnout_risk	Classification target

training and 20% for testing. This produced 1,440 training rows and 360 test rows, with zero overlap between train and test users. This is more realistic than a random row split because the model is evaluated on unseen users.

Categorical variables were encoded with one-hot encoding and numeric variables were standardized with StandardScaler. All transformations were placed inside a ColumnTransformer and a scikit-learn Pipeline; thus, encoders and scalers are fitted only on training data.

Several interpretable engineered features were added: work_intensity, screen_work_ratio, meeting_density, sleep_deficit and overwork_flag. These variables express workload pressure, screen exposure, meeting concentration and sleep deficit.

The preprocessing pipeline also improves reproducibility. Every candidate model receives exactly the same transformed feature matrix, and the same logic is available at inference time if the fitted pipeline is saved. This is preferable to manually transforming training and test sets in separate notebook cells, which would make leakage and inconsistent preprocessing more likely.

C. Modeling Strategy

The primary task is regression on burnout_score. The secondary task is classification on burnout_risk. This choice is important because the categorical target is highly imbalanced: 1,527 rows are Low, 253 are Medium, and only 20 are High. Therefore, a classifier can obtain high accuracy while failing on the most critical cases.

For regression, the following models were evaluated: DummyRegressor, Ridge Regression, K-Nearest Neighbors, Random Forest and Hist Gradient Boosting. For classification, the models were DummyClassifier, balanced Logistic Regression, balanced Random Forest and Hist Gradient Boosting. Cross-validation used GroupKFold on the training set. A light hyperparameter search was then performed for Random Forest regression with RandomizedSearchCV.

The main regression metric is mean absolute error (MAE), supported by RMSE and R^2. The classification metrics are macro F1, balanced accuracy and class-level precision/recall. These metrics are aligned with the course recommendation to define clear, task-dependent evaluation criteria rather than optimizing accuracy blindly.

The selected algorithms cover different inductive biases. Ridge Regression is a linear baseline with regularization and is useful to check whether the target is mostly explained by linear relationships. KNN is sensitive to scaling and local neighborhood structure. Random Forest and Hist Gradient Boosting can capture non-linear effects and interactions. Dummy models provide lower bounds: if a candidate does not clearly beat them, it should not be considered useful.

Hyperparameter optimization was deliberately limited. The course material on AutoML emphasizes that automated search is useful, but it should not replace problem understanding. Therefore, only a small randomized search was applied to Random Forest after comparing the initial candidates. The final test set was not used to choose hyperparameters.

III. RESULTS

A. Data Insights

The data understanding phase showed that task_completion_rate is the dominant feature: it has a correlation of about -0.96 with burnout_score. This is predictive, but it must be interpreted with caution. It may be a direct proxy for burnout or part of the synthetic logic used to generate the target. A univariate Ridge model fitted on task_completion_rate alone achieves test MAE 4.83 and $R^2 = 0.934$, nearly matching the full thirteen-feature model ($R^2 = 0.936$); the remaining features therefore contribute almost no incremental predictive power, which makes the case for careful external validation before any deployment. Other variables such as work_hours, screen_time_hours and meetings_count have weaker positive correlations with burnout.

The descriptive analysis also confirmed that the dataset is clean in a technical sense: there are no missing values, no duplicated rows, and each user has exactly 10 observations. However, technical cleanliness is not the same as data quality for decision making. Since the dataset is public and its collection context is limited, representativeness cannot be guaranteed. This limits the strength of any claim about real organizations.

An explicit data quality report was added to the notebook, including checks for invalid ranges, duplicates and missing values. Statistical outliers were also detected through the IQR method for the main numeric variables. They were not removed automatically: in this domain, extreme burnout scores or extreme working patterns may be the most relevant observations rather than errors.

The risk-label distribution is a major issue. The High class represents only about 1.1% of the dataset. This affects model selection, evaluation and fairness: the most important class from a preventive perspective is also the least represented.

A programmatic check confirmed that the three risk bands occupy strictly non-overlapping intervals of burnout_score (Low: $\leq$69.92, Medium: 70.01–109.99, High: $\geq$110.22), meaning that burnout_risk is a deterministic bucketisation of `burnout_score` rather than an independent label. A regression model that predicts the score accurately can therefore be thresholded into a near-perfect classifier; the value of the classification task lies in interpretability and operational communication rather than in genuinely new predictive signal.

For this reason, the project treats `burnout_risk` as a secondary task. The continuous score gives a more stable learning signal, while the classes are useful to communicate results and inspect whether high-risk observations are missed.

B. Regression Results

Table III reports the test-set results for the main regression models. All meaningful models outperform the dummy baseline by a large margin. Ridge obtains the lowest raw test MAE among the initial models, while Random Forest has the best cross-validation MAE and was selected for tuning.

TABLE III
REGRESSION PERFORMANCE ON THE TEST SET.

Model	MAE	RMSE	R^2
Ridge	4.79	6.31	0.934
Random Forest	4.85	6.41	0.932
Hist Gradient Boosting	5.14	6.73	0.925
KNN	9.26	11.65	0.776
Dummy median	20.35	25.80	-0.099

After tuning, Random Forest used 200 estimators, maximum depth 4, `min_samples_leaf=8` and `max_features=1.0`. The tuned model obtained CV MAE 4.85, test MAE 4.88, test RMSE 6.21 and test $R^2 = 0.936$. This is a strong improvement over the dummy baseline and confirms that the selected features contain useful predictive information.

It is worth noting that the plain untuned Ridge achieves a lower raw test-set MAE (4.79) than the tuned Random Forest (4.88). Random Forest was retained as the primary model because it outperforms Ridge on cross-validated MAE (4.97 vs. 5.06), which provides a more reliable generalisation estimate than a single held-out draw. The difference on the test set is within typical sampling noise for 360 observations, so neither direction is conclusive; the CV result is the appropriate selection criterion and the choice is explicitly stated here rather than left implicit.

Cross-validation on the training set supported the same conclusion. Random Forest obtained mean CV MAE 4.97, Ridge 5.06, Hist Gradient Boosting 5.19, KNN 9.03 and the dummy baseline 18.46. The gap between training and test error was monitored to detect overfitting: tree-based models obtained lower training error but did not collapse on unseen users, while KNN remained substantially weaker.

The error analysis in the notebook includes a predicted-versus-actual plot, the distribution of residuals and the largest individual errors. These checks are important because an average error can hide systematic failures. In this domain, large underestimation for high-burnout records would be more problematic than small errors around low-risk observations.

To address a key methodological risk, a sensitivity analysis was performed by removing `task_completion_rate`. Results dropped sharply: Ridge reached MAE 20.25 and Random Forest MAE 20.26, close to the dummy baseline. This confirms that `task_completion_rate` carries most of the predictive signal. The model is therefore useful as a predictive pipeline, but it should not be presented as evidence of causal burnout mechanisms.

C. Classification Results

The best classifier was balanced Logistic Regression, with balanced accuracy 0.803 and macro F1 0.646. Table IV reports its class-level metrics. The model performs very well on `Low` and reasonably on `Medium`; the `High` class remains difficult because only four high-risk examples are present in the test set.

TABLE IV
CLASSIFICATION REPORT FOR BALANCED LOGISTIC REGRESSION.

Class	Precision	Recall	F1	Support
Low	0.99	0.93	0.96	294
Medium	0.68	0.73	0.70	62
High	0.17	0.75	0.27	4

The low precision on `High` indicates many false alarms, while the recall of 0.75 suggests that most observed high-risk test cases were detected. In a preventive setting, this trade-off may be acceptable only if predictions are reviewed by humans and not used automatically.

The classification results also show why accuracy alone would be misleading. The dummy classifier that always predicts the majority class is not useful, even though most observations are `Low`. Macro F1 and balanced accuracy give a more honest view because they weight minority classes more appropriately.

D. Interpretability

Permutation importance was computed on the regression test set. The most important feature is `task_completion_rate`. This is consistent with the correlation analysis, but it should not be interpreted causally. A low task completion rate may be a consequence, a symptom, a proxy, or a construction component of burnout score. Other features, such as workload and screen time, provide weaker but more directly actionable information. This distinction is important for business interpretation: interventions should not simply pressure employees to complete more tasks, because that may worsen the underlying problem.

E. Fairness and Ethics

The dataset does not include sensitive attributes such as gender, age or ethnic origin. Nevertheless, fairness cannot be ignored. The notebook checks regression errors and high-risk recall across operational subgroups such as `day_type`, `after_hours_work` and work-hour bands. These checks are not a full fairness audit, but they are useful to detect systematic performance differences.

The per-subgroup High-risk recall figures must be interpreted with caution. The test set contains only four High-risk cases in total, distributed across multiple subgroups, so individual groups have between zero and four positive instances. A single prediction can swing recall between 0.0 and 1.0, and a NaN appears for groups with no positive cases at all. These values carry no statistical weight and should be read only as a directional sanity check, not as evidence of systemic (un)fairness in either direction.

From an ethical standpoint, the safest deployment scenario is aggregate monitoring. For example, the model could help identify teams or periods where workload patterns become riskier. Individual predictions should require human review, transparency and consent. The model should never be used as a punitive tool or as an automatic performance evaluation system.

IV. Conclusions

The project implements the required end-to-end machine learning workflow: problem definition, public dataset selection, preprocessing, modeling, evaluation, insight extraction and deployment discussion. The regression task is the most reliable formulation: the tuned Random Forest reaches MAE 4.88 and $R^2 = 0.936$ on unseen users, clearly outperforming the baseline. The classification task is useful for operational interpretation, but the `High` class is too rare for strong claims.

Only the regression pipeline is persisted. Because `burnout_risk` is a deterministic function of `burnout_score` (verified in the notebook, Section 3.1), any downstream system can derive the risk label from the regression output by applying fixed thresholds; saving a separate classification pipeline would therefore duplicate functionality without adding value.

The most important insight is that `task_completion_rate` dominates the prediction. This improves performance but also creates a methodological risk: the variable may be too close to the target-generation process. Future work should validate the approach on an external real-world dataset, collect richer contextual information, define intervention thresholds with domain experts and run a stronger fairness assessment before any deployment.

References

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